Cohera: Why quantum magnetic navigation, why Ireland, why now

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Each motivation follows the same pattern: the operational challenge, why GNSS fails there, the impact on uncrewed missions, and how MagNav helps. Every factual claim links to its source.

1 Ireland's problem: a huge sea, a small fleet, critical cables

- **Ireland's sea area is about 880,000 km^2, more than 10× its land area.** Patrolling it with crewed ships and aircraft is not feasible; persistent uncrewed patrol is the only way to scale. [Marine Institute]
- **About 75% of transatlantic subsea cables pass through or near Irish waters.** The Naval Service has 8 ships but often only 1–4 can sail because of crew shortages, and Ireland relied on the UK to track a Russian submarine off Cork. [TechCentral, Jul 2026]
- **Ireland does not know who is in its waters.** The Wall Street Journal's video on Ireland's exposure to Russian activity. [WSJ video] *(TODO: confirm title and date)*; see also [Braw in WSJ, Atlantic Council]
- **The Russian ship *Yantar***, believed capable of interfering with cables and of carrying underwater drones, was escorted out of the Irish Sea in Nov 2024. It returned towards UK and Irish waters in Nov 2025 with its identification transponder (AIS) switched off. [Afloat, Nov 2025]
- **Four unidentified military-specification drones** entered Dublin Airport's no-fly zone during President Zelenskyy's arrival (1 Dec 2025), then circled an Irish Navy ship; a launch from an undetected vessel was being investigated. [NV / The Journal, Dec 2025]
- **Ireland has had no long-range primary military radar.** The first systems are due in 2027, with full capability by the end of 2028 (about EUR 500M, with France). Until then, aircraft and drones with transponders off are hard to see. [Irish Times, Jul 2026]
- **The gap, in one sentence:** Ireland needs cheap, persistent uncrewed surveillance over sea, and that is exactly where an adversary will jam GNSS.

2 GNSS denial is already happening, including over Irish airspace

- **About 123,000 flights were affected by GNSS interference over the Baltic in four months** (Jan–Apr 2025), according to a report to ICAO. [Travel Tomorrow / SVT]
- **The European Commission President's plane lost GPS approaching Bulgaria** (Sep 2025), with Russian jamming suspected. [AP, Sep 2025]
- **More than 2,300 Ryanair flights reported GPS interference** (Aug 2023 – May 2024). [Irish Times, May 2024]
- **Ireland is keeping old ground beacons** because of jamming. IATA reports jamming up 67% and spoofing up 193% (2023–2025), with an estimated 1,500 flights a day affected. Interference has occurred in Irish-controlled Atlantic airspace. [Irish Times, Jun 2026]

- **Impact on drones:** a small drone has no ground beacons and no strategic-grade inertial system. When GNSS goes, its consumer-grade inertial sensors drift by hundreds of metres within minutes, so the mission is lost or the drone is captured.

3 Why the sea defeats every other GNSS backup

- **Cameras:** the sea surface has no fixed features and looks the same everywhere, so visual odometry and scene matching have nothing to lock on to. They also fail at night and in fog.
- **Lidar and radar altimetry:** the surface moves and is flat, so terrain-contour matching (TERCOM-style) has no terrain. Both also *emit*, which gives away the platform's position.
- **Star trackers:** need a clear sky, which Ireland's low cloud and frequent cloud cover rarely give.
- **Inertial only:** the error grows without bound, roughly as $\frac{1}{2}bt^2$ from accelerometer bias and $g\omega t^3/6$ from gyro bias.
- **The magnetic field:** the crustal anomaly field is fixed to the rock beneath the surface, does not care about waves, weather or daylight, and needs no line of sight.
- **Industry confirms it:** Northrop Grumman and SandboxAQ say their magnetic navigation works "over open water and feature-limited terrain, where visual navigation struggles". [The Aviationist, Oct 2026]

4 Why Ireland, why now: the map exists

- **The Tellus airborne survey flew its final flight in June 2026**, after 16 years and more than 440,000 km of flight lines measuring the magnetic field, electromagnetics and radioactivity across the whole island. Ireland is "one of the few countries in the world" with a completed detailed national airborne geophysics survey. About 80% of the processed data is already free to download, and the merged national dataset is due in 2027. [gov.ie, Jun 2026]
- **The data is open (CC BY 4.0)**, at a 60 m survey height and 200 m line spacing. That resolution is far better than most national magnetic maps. A navigation map normally costs a survey campaign; Ireland's is free and already flown.
- **Ground observatories for day-to-day field variation:** MagIE (Dunsink, Birr, Armagh, Valentia) records the daily geomagnetic variation, which we subtract from readings so they match the map.
- **Honest gap and opportunity:** offshore coverage relies on coarser global grids such as [EMAG2]. Extending Tellus offshore, as INFOMAR did for the seabed, would give Ireland a complete magnetic navigation map of its whole marine area.

5 Why quantum (OPM) and why it is new

- **Miniature optically pumped magnetometers (OPMs) are recent.** The chip-scale atomic magnetometer was first shown at NIST in the mid-2000s. [NIST] Total-field units small enough for small drones have only recently become commercial. [QuSpin QTFM Gen-2]
- **Size, weight and power fit a small drone:** QTFM Gen-2 is 41 g and about 2.5 W, with $3\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$ typical sensitivity, under 3 nT heading error, a dead zone under $\pm 7°$, and up to 1 kHz sampling. Classic survey caesium magnetometers weigh kilograms. [QuSpin]
- **It measures the field as a frequency.** Rubidium atoms precess at about 7 Hz per nT, a constant of nature, so there is no scale drift and no calibration decay. A MEMS or fluxgate compass has bias, scale and misalignment errors that, on a 49,000 nT field, become **50–500 nT** of attitude-dependent error. That is larger than the map signal itself.

- **Our own data shows this:** the vector channel's magnitude wandered 884 nT, against 79 nT for the scalar OPM, on the same flight. Our X500's built-in compasses wander 400–1000 nT in flight. Tellus anomalies are tens to hundreds of nT. *(Figures from the repo notebooks.)*
- **Signal-to-noise:** at $3\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$ against 10–1000 nT anomalies, the sensor itself is not the limit; platform interference and map quality are. That is where we put our effort.

6 Why you can't jam it from a distance (headline figures)

The jammer has to create a magnetic *dipole moment* m, measured in $\mathrm{A{\cdot}m^2}$ (current × loop area). Its field falls off as the **cube** of distance, while a radio jammer's power falls off only as the square. On the dipole axis (the jammer's best case):

$$B(r) = \frac{\mu_0\, m}{2\pi r^3} = 2\times 10^{-7}\,\frac{m}{r^3}\ \mathrm{T} \qquad \Rightarrow \qquad m_{\mathrm{needed}} = \frac{r^3\,\Delta B}{2\times 10^{-7}}.$$

To shift the field by 10 nT at…	100 m	1 km	10 km
Dipole moment needed	$5\times10^{4}\,\mathrm{A{\cdot}m^2}$	$5\times10^{7}\,\mathrm{A{\cdot}m^2}$	$5\times10^{10}\,\mathrm{A{\cdot}m^2}$
DC power, 20 m copper loop, 5 t	∼3 W	∼3 MW	∼3 TW
Permanent magnet (NdFeB)	$\sim$0.05 $\mathrm{m^3}$	$\sim$50 $\mathrm{m^3}$ (∼360 t)	$\sim$48,000 $\mathrm{m^3}$

- **Power for a resistive coil:** $P = 4\rho_{\mathrm{Cu}}\, d_{\mathrm{Cu}}\, m^2/(a^2 M)$, where a is the loop radius and M the copper mass. Power grows with m^2, so it grows with r^6: **each 10× in range costs 1,000,000× in power.**
- **3 TW is about the average output of every power station on Earth combined**, just to nudge the field by 10 nT at 10 km.
- **Close range is the only real threat:** $1\,\mathrm{A{\cdot}m^2}$ (a hand-sized coil at a few amps) gives 200 nT at 1 m, 0.2 nT at 10 m, 0.2 pT at 100 m and 0.2 *femto*tesla at 1 km. A jammer would have to be within tens of metres. That is the same platform-interference problem we already measure and compensate.
- **A jammer gives itself away:** a strong artificial dipole is a magnetic beacon that the same sensors detect, as in magnetic anomaly detection (MAD) of submarines. The filter's integrity check also flags readings that don't match the map.
- **Compare GNSS:** the GPS L1 signal reaches the ground at only about −158.5 dBW ($\approx 1.4\times10^{-16}$ W). [IS-GPS-200] A **1 W** jammer 10 km away, in line of sight, overpowers it by about **42 dB (16,000×)**, and still by about 22 dB at 100 km.
- **Headline:** *about 1 W denies GPS at 10 km; about 3 TW is needed to disturb MagNav at 10 km.*
- **Spoofing:** a GNSS spoofer only has to broadcast a fake signal. To spoof MagNav, an adversary would have to rebuild the rock beneath the sea.

7 Operational advantages

- **Passive:** it emits nothing, so it can't be detected or direction-found. Lidar, radar altimeters and datalinks all emit.
- **Bounded error:** it corrects inertial drift continuously, instead of the error growing with time.
- **Works in any weather and at any time:** night, fog, cloud, sea state.
- **No infrastructure:** no satellites, beacons or ground stations to attack.

- **Low cost per platform:** a gram-scale sensor plus an open map, instead of a strategic-grade inertial system.
- **One sensor, many platforms:** the same sensor and software work on multirotors for coastal patrol and fixed wings for long offshore legs. Our build shows both (X500 and Moose).
- **Dual use:** the same platform does mineral exploration, UXO survey and cable-route inspection in peacetime.

8 Who else is doing this (2025–2026)

- **Northrop Grumman and SandboxAQ (Sep 2026):** quantum magnetic navigation flown on the Lumberjack Group-3 uncrewed aircraft. Reported as resisting jamming and working over open water. [Northrop Grumman]
- **Honeywell and DIU (Sep 2026):** an Embraer 170 flew 4 h 23 min over the Pacific without GPS, with 89% better position accuracy than conventional backups. [DefenseScoop]
- **Q-CTRL (2025):** airborne quantum MagNav errors up to about 46× lower than an aided strategic-grade inertial system; best final error 22 m (0.006% of distance flown). [arXiv:2504.08167]
- **DAF–MIT MagNav challenge:** a public benchmark with real flight data and maps. [magnav.mit.edu], [arXiv:2007.12158]
- **The European gap:** the leading demonstrations are US and Australian. Ireland has the map, the urgent need and the talent. Cohera is a European, open-map, small-drone version.

9 What we still have to prove (judges will ask)

- **Platform interference** from motors and wiring: our bench tests measure it and our mounting reduces it.
- **Flat-map regions and offshore map resolution**, addressed by our navigability and integrity map.
- **Geomagnetic storms**, addressed with MagIE corrections and integrity monitoring.
- **Export control:** the QTFM is ECCN 6A006.a.2, while the QTFM-B ($>20\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$) is 6A996.a. Our breaking-point sweep tests whether the lower-sensitivity, easier-to-export unit is enough.

Contains Irish Public Sector Data (Geological Survey Ireland) licensed under CC BY 4.0. Jamming figures are first-principles estimates (on-axis dipole, DC, copper at 20 °C, NdFeB $B_r \approx 1.3\,\mathrm{T}$, free-space path loss at 1575.42 MHz); they are orders of magnitude, not a threat assessment.